

Integration by Parts

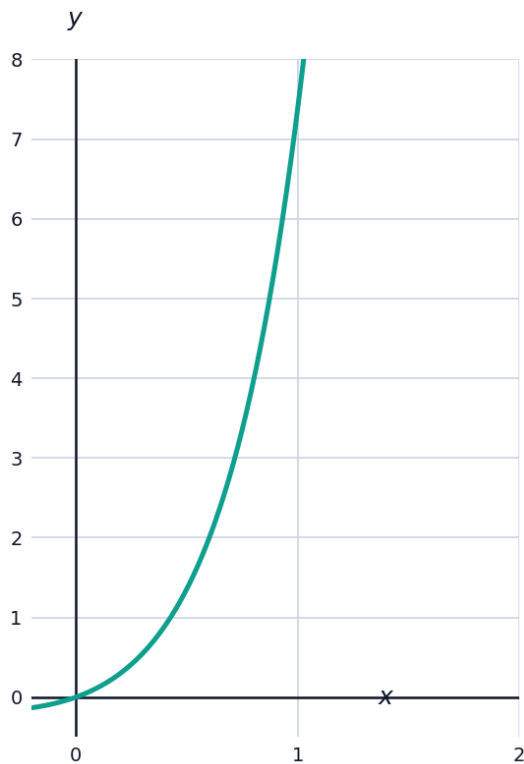


The integration by parts formula

- 1 Evaluate $\int x e^x dx$ using integration by parts.
- 2 Evaluate $\int x \cos x dx$ using integration by parts.
- 3 Evaluate $\int \ln x dx$ using integration by parts (with $dv = dx$).

Repeated and definite applications

- 4 Evaluate $\int x^2 e^x dx$, applying integration by parts twice.
- 5 Evaluate $\int_0^1 x e^{2x} dx$ exactly.



Choosing u and dv wisely (LIATE)

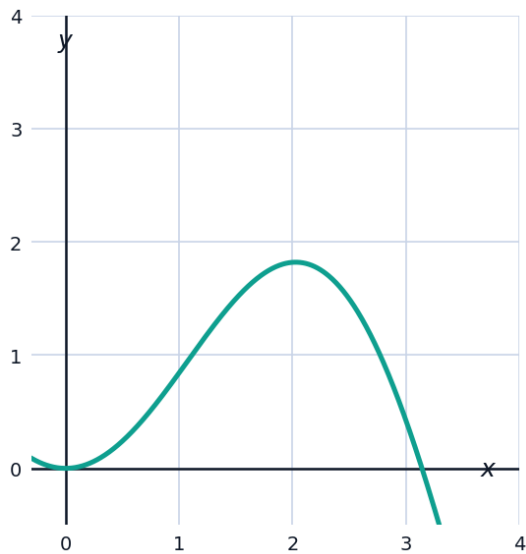
- 6 Evaluate $\int x \ln x dx$, choosing $u = \ln x$ (logarithmic functions are prioritized as u under the LIATE rule).
- 7 Evaluate $\int x^2 \cos x dx$, applying integration by parts twice.

The recurring-integral trick

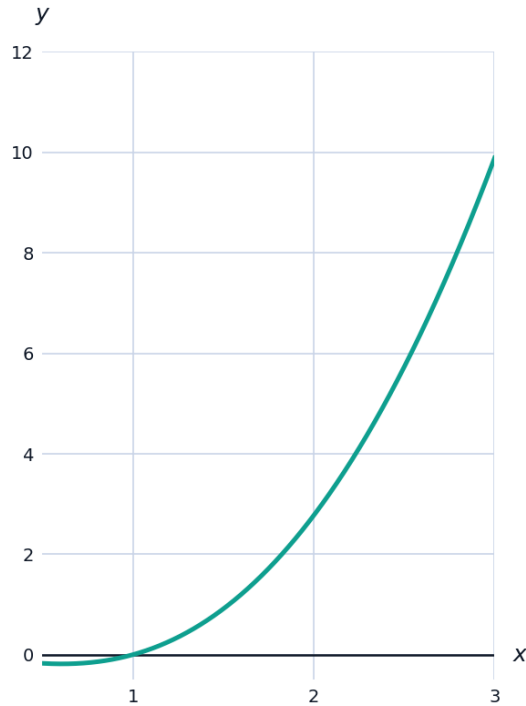
- 8 Evaluate $\int e^x \sin x \, dx$ using the technique of applying integration by parts twice and solving algebraically for the original integral.
-

Definite integrals by parts

- 9 Evaluate $\int_0^\pi x \sin x \, dx$ exactly.



- 10 Evaluate $\int_1^e x^2 \ln x \, dx$ exactly.



Reduction formulas

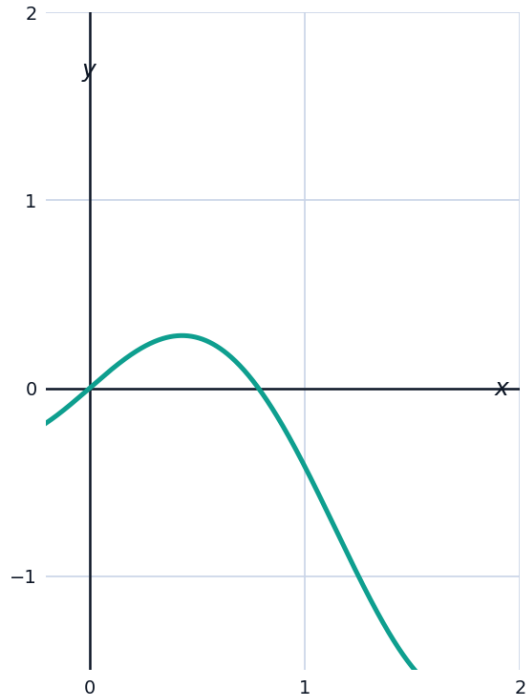
- 11 Use integration by parts with $u = x^n$, $dv = e^x dx$ to derive the reduction formula $\int x^n e^x dx = x^n e^x - n \int x^{n-1} e^x dx$.
- 12 Use the reduction formula above (applied twice) to evaluate $\int x^2 e^x dx$, confirming it matches an earlier result.

Integration by parts with inverse trig functions

- 13 Evaluate $\int \tan^{-1} x \, dx$ using integration by parts with $u = \tan^{-1} x$, $dv = dx$.
- 14 Evaluate $\int x \sin^{-1} x \, dx$ using $u = \sin^{-1} x$, $dv = x \, dx$ (the resulting integral can be left in terms of a known antiderivative form).

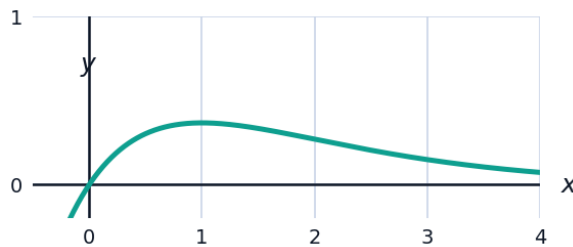
A capstone mixed-technique problem

- 15 Evaluate $\int_0^{\pi/2} x \cos(2x) dx$ exactly, using integration by parts.



Visualizing an area found by integration by parts

- 16 The graph shows $f(x) = xe^{-x}$ on $[0, 3]$. Find the exact area under the curve on this interval using integration by parts.



Choosing between substitution and parts

- 17 For each integral, state whether substitution or integration by parts is the appropriate technique, and identify the key choice (u -substitution variable, or u/dv for parts) without fully evaluating: (a) $\int xe^{x^2} dx$; (b) $\int xe^{2x} dx$.
- 18 Evaluate $\int \sec^3 x dx$ using integration by parts with $u = \sec x$, $dv = \sec^2 x dx$, and the identity $\tan^2 x = \sec^2 x - 1$ to solve the resulting equation for the original integral.

- 19 A city's population growth rate is modeled by $r(t) = t \ln(t + 1)$ thousand people per year, for $0 \leq t \leq 5$. Set up and evaluate the definite integral for the total population growth over these 5 years using integration by parts, giving your answer to two decimal places.

